NIH BIOGRAPHICAL SKETCH COMMON FORM

Name: Rangaraju, Vidhya

Persistent Identifier (PID) of the Senior/Key Person: https://orcid.org/0000-0002-6716-4312

Position Title: Research Group Leader

Organization and Location: Max Planck Florida Institute for Neuroscience, Jupiter, Florida, United States

PROFESSIONAL PREPARATION

INSTITUTION AND LOCATION	DEGREE	Start Date	Completion Date	FIELD OF STUDY
Max Planck Institute for Brain Research, Frankfurt, Not Applicable, N/A, Germany	Postdoctoral Fellow	02/2014	12/2019	Neuroscience
Weill Cornell Medicine of Cornell University, NY, NY, United States	PhD	07/2007	01/2014	Chemical Biology and Neuroscience
Anna University, Chennai, Not Applicable, N/A, India	Other Baccalaureate (BOTH)	07/2002	05/2006	Industrial Biotechnology

Appointments and Positions

2020 - present	Research Group Leader, Max Planck Florida Institute for Neuroscience, Jupiter, Florida, United States
2020 - present	Affiliate Faculty, Charles E. Schmidt College of Science, Florida Atlantic University, Jupiter, Florida, United States
2014 - 2019	Postdoctoral Fellow, Max Planck Institute for Brain Research, Synaptic Plasticity, Frankfurt, Not Applicable, N/A, Germany
2007 - 2014	Tri-Institutional Chemical Biology Graduate Student, Weill Cornell Medicine of Cornell University, New York, New York, United States

Products

Products Closely Related to the Proposed Project

1. Rangaraju V, Calloway N, Ryan TA. Activity-driven local ATP synthesis is required for synaptic function. Cell. 2014 Feb 13;156(4):825-35. PubMed Central PMCID: PMC3955179.
2. Rangaraju V, Lauterbach M, Schuman EM. Spatially Stable Mitochondrial Compartments Fuel Local Translation during Plasticity. Cell. 2019 Jan 10;176(1-2):73-84.e15. PubMed PMID: 30612742.
3. Bapat O, Purimetla T, Kruessel S, Shah M, Fan R, Thum C, Rupprecht F, Langer JD, Rangaraju V. VAP spatially stabilizes dendritic mitochondria to locally support synaptic plasticity. Nat Commun. 2024 Jan 4;15(1):205. PubMed Central PMCID: PMC10766606.
4. Ghosh I, Fan R, Shah M, Bapat O, Rangaraju V. Synapses drive local mitochondrial ATP synthesis to fuel plasticity. bioRxiv [Preprint]. 2025 April 10. Available from: https://www.biorxiv.org/content/10.1101/2025.04.09.648032v1 DOI: 10.1101/2025.04.09.648032
5. Shah M, Ghosh I, Shree Ramesh N, Pishos L, Pancani T, Villani V, Yasuda R, Sun C, Kamasawa N, Rangaraju V. Mitochondria structurally remodel near synapses to fuel the sustained energy demands of plasticity. bioRxiv : the preprint server for biology. 2026 July 27. Available from: https://www.biorxiv.org/content/10.1101/2025.08.27.672715v2

Other Significant Products Highlighting Contributions to Science

1. Sun C, Nold A, Fusco CM, Rangaraju V, Tchumatchenko T, Heilemann M, Schuman EM. The prevalence and specificity of local protein synthesis during neuronal synaptic plasticity. Sci Adv. 2021 Sep 17;7(38):eabj0790. PubMed Central PMCID: PMC8448450.
2. Amrapali Vishwanath A, Comyn T, Mira RG, Brossier C, Pascual-Caro C, Faour M, Boumendil K, Chintaluri C, Ramon-Duaso C, Fan R, Ghosh K, Farrants H, Berwick JP, Sivakumar R, Lopez-Manzaneda M, Schreiter ER, Preat T, Vogels TP, Rangaraju V, Busquets-Garcia A, Plaçais PY, Pavlowsky A, de Juan-Sanz J. Mitochondrial Ca(2+) efflux controls neuronal

metabolism and long-term memory across species. Nat Metab. 2026 Feb;8(2):467-488. PubMed Central PMCID: PMC12945686.

3. Benedetti L, Fan R, Weigel AV, Moore AS, Houlihan PR, Kittisopikul M, Park G, Petruncio A, Hubbard PM, Pang S, Xu CS, Hess HF, Saalfeld S, Rangaraju V, Clapham DE, De Camilli P, Ryan TA, Lippincott-Schwartz J. Periodic ER-plasma membrane junctions support long-range Ca(2+) signal integration in dendrites. Cell. 2025 Jan 23;188(2):484-500.e22. PubMed PMID: 39708809.
4. Moret A, Farrants H, Fan R, Zingg K, Gee C, Oertner T, Rangaraju V, Schreiter E, de Juan-Sanz J. An expanded palette of bright and photostable organellar Ca^{2+} sensors. [Preprint]. 2025 January 12. DOI: 10.1101/2025.01.10.632364
5. Heumüller M, Glock C, Rangaraju V, Biever A, Schuman E. A genetically encodable cell-type-specific protein synthesis inhibitor. Nature Methods. 2019; 16(8):699-702. Available from: https://www.nature.com/articles/s41592-019-0468-x DOI: 10.1038/s41592-019-0468-x

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

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Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§287, 1001, 1031 and 31 U.S.C. §§3729-3733 and 3802.

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NIH BIOGRAPHICAL SKETCH SUPPLEMENT

Name: Rangaraju, Vidhya

Persistent Identifier (PID) of the Senior/Key Person: https://orcid.org/0000-0002-6716-4312

Position Title: Research Group Leader

Organization and Location: Max Planck Florida Institute for Neuroscience, Jupiter, Florida, United States

Personal Statement

I am a Research Group Leader at the Max Planck Florida Institute for Neuroscience (MPFI). My lab seeks to bridge the knowledge gap between mitochondria and memory formation. Our findings indicate that mitochondria undergo structural remodeling and synthesize energy on demand in response to synaptic plasticity (Rangaraju et al. 2014, Ghosh et al. 2025, Shah et al. 2026). We have also shown that mitochondria are locally stabilized near synapses to meet the immediate and local energy demands of synaptic plasticity (Rangaraju et al. 2019, Bapat et al. 2024, Ghosh et al. 2025, Shah et al. 2026). We employ several cutting-edge approaches, including subcellular proteomics, state-of-the-art ATP and calcium imaging in the spine and mitochondria, and various genetic and biochemical tools to identify the fundamental mitochondrial mechanisms that drive synaptic plasticity.

I have mentored several trainees over my 6.5-year career as an independent investigator, all of whom have been successful in their academic pursuits (4 postbaccalaureate and 3 undergraduate alumni pursuing PhD and MD in top schools). I lead an eight-member research team (4 postdoctoral fellows – 1 completing in 2027, 2 graduate students – completing in 2027 and 2028, 1 staff scientist, and 1 lab manager). Our team is multidisciplinary, with a skill set spanning cellular and molecular biology, super-resolution and electron microscopy, advanced proteomics, in vivo imaging, and animal behavior in rodent, human, and disease models. Our group is ideally positioned with the wealth of additional expertise at the MPFI.

During my Ph.D. in Timothy Ryan's lab at Weill Cornell Medicine, I developed and employed a novel luciferase-based ATP reporter (U.S. Patent 9487819 B2) and demonstrated that ATP is locally produced at synaptic terminals in response to neuronal activity. Even brief perturbations in activity-driven ATP synthesis dramatically affected the synaptic vesicle cycling pathway and subsequent synaptic communication (Rangaraju et al. 2014). As such neuronal defects have been observed in many neurological disorders, these results emphasize the importance of synthesizing ATP on demand to support synaptic function.

As a postdoc in Erin Schuman's lab at the Max Planck Institute for Brain Research in Germany, I developed novel approaches to visualize newly synthesized proteins in spines using two-photon glutamate uncaging, locally manipulate mitochondrial function via optogenetics, and image live mitochondria using super-resolution. As a result, I showed that mitochondria form stable compartments that serve as the local energy supply required for protein synthesis and memory formation, the absence of which could result in memory impairment (Rangaraju et al. 2019).

Consistent with my track record of tackling uncharted research questions with novel approaches, we continue to address our research questions rigorously, as outlined in the 'Products' section, which highlights my expertise in the methods and questions we pursue in our research program.

Honors

2025	McKnight Scholars Program Finalist, McKnight Foundation
2024	Janett Rosenberg Trubatch Career Development Award, Society for Neuroscience
2024 - 2029	NIH Director's New Innovator Award (DP2) , National Institutes of Health
2024 - 2025	CZI Collaboration Supplement, Chan Zuckerberg Initiative
2023	Freeman Hrabowski Scholars Program Finalist, Howard Hughes Medical Institute
2023 - 2027	Ben Barres Early Career Acceleration Award, Chan Zuckerberg Initiative
2020	Peter and Patricia Gruber International Research Award, Society for Neuroscience
2019	Scientific Discovery of the Year, Max Planck Institute for Brain Research
2019	MacGillavry Fellowship, University of Amsterdam
2018	Lindau Nobel Laureate Meeting Award, Lindau Nobel Laureate Meetings
2016 - 2017	Marie Curie Actions Individual Postdoctoral Fellowship, Marie Skłodowska-Curie Actions

2015	EMBO Long Term Postdoctoral Fellowship, European Molecular Biology Organization
2015	Humboldt Research Fellowship for Postdoctoral Researchers, Humboldt Foundation
2012	Vincent du Vigneaud Award of Excellence, Weill Cornell Medicine of Cornell University

Contributions to Science

1. Identified the energy consumers and the mechanisms of energy synthesis in presynaptic terminals:
The brain consumes 20% of the human body's total energy to perform critical functions, including learning and memory formation. While neuronal synapses are hotspots of energy consumption, the locus of metabolic control was unknown. To address this question, I developed an optical reporter (U.S. Patent, Rangaraju et al. 2014) to visualize and measure changes in ATP levels in individual living nerve terminals. In addition, I custom-built a highly sensitive microscope, as conventional microscopes could not detect the luminescence signal produced by luciferase. Using these novel approaches, I found that ATP is locally produced at synaptic terminals in response to neuronal activity to fuel synaptic transmission. Furthermore, I genetically disrupted the synaptic vesicle cycling pathway and showed that synaptic vesicle cycling consumes the most energy in nerve terminals. This research, published in Cell, shed new light on the molecular balance between electrical activity and synaptic energy synthesis (Rangaraju et al. 2014). Our findings established a new paradigm for studying the metabolic control of cognition and neurodegeneration (Yang et al. 2024), shifted the research direction of the Ryan Lab (Ph.D. lab), and inspired me to establish my own lab to study Neuroenergetics.

2. Discovered the local mechanisms of energy synthesis in postsynaptic spines and dendrites:
Memory formation is highly energy-consuming, but its energy supply was unknown. I addressed this question in my postdoctoral work by developing tools to label newly synthesized proteins in spines following spine stimulation, locally disrupt mitochondrial function via optogenetics, and resolve live mitochondrial structures using super-resolution microscopy. Using these new approaches, I showed that locally stable mitochondria fuel the protein synthesis required for synaptic plasticity, a cellular correlate of memory formation, and published the findings in Cell (Rangaraju et al. 2019). Using an unbiased proteomic screening approach, we identified VAP (Vesicle-Associated Membrane protein-associated Protein), a protein implicated in amyotrophic lateral sclerosis, as a molecular glue that stabilizes mitochondria by tethering them to actin, as reported in Nature Communications (Bapat et al. 2024). We developed novel imaging tools and strategies to measure ATP within individual spines and mitochondria, demonstrating that locally stable mitochondria near dendritic spines generate instantaneous and sustained ATP to support synaptic plasticity (Ghosh et al. 2025). We recently developed a CLEM/ET pipeline along with machine learning-based EM image segmentation and quantification to show that the locally stable mitochondria near spines undergo substantial structural remodeling in their inner cristae structure, energy synthesis machinery, and their associations with endoplasmic reticulum and ribosomes to generate the ATP needed to support synaptic plasticity (Shah et al. 2026).

3. Characterized the cellular mechanisms of dendrites during synaptic plasticity:
As neurons have a complex morphology, it was unclear how they accomplish the large-scale resource allocation of proteins and RNAs to meet local demands during synaptic plasticity, given a fixed global budget for protein production. It was unclear how local spine stimulation affects the size of translation compartments, including the spatial spread of nascent protein and RNA. We used metabolic labeling and DNA-PAINT to visualize and quantify nascent proteins during synaptic plasticity. We discovered a local 'neighborhood' of synapses that are supplied with newly synthesized proteins during synaptic plasticity, as reported in Science Advances (Sun et al. 2021). We also showed that long non-coding RNAs are recruited to plasticity-induced spines to mediate structural plasticity and fear memory consolidation, a finding published in Nature Communications (Espadas et al. 2024). With our ongoing interest in unraveling the significance of the mitochondria and endoplasmic reticulum in cellular mechanisms within dendrites, we showed that: (i) ER forms a ladder-like array along dendrites to enable long-range calcium propagation important for dendritic computations and synaptic plasticity (Benedetti et al. 2025); (ii) as mitochondria can serve as calcium regulators, we showed the significance of mitochondria in maintaining spine and dendritic calcium to maintain long-term memory using state-of-the-art mitochondrial calcium imaging (Vishwanath et al. 2026).

4. Developed genetically encoded tools to probe energy synthesis and protein synthesis mechanisms in neurons:
   I have actively contributed to neuroscience by developing novel tools to dissect different mechanisms of neuronal biology. I designed, optimized, and characterized a novel ATP reporter based on the firefly enzyme luciferase to quantify ATP concentrations in living nerve terminals (Rangaraju et al. 2014, Ghosh et al. 2025). This work influenced my Ph.D. mentor, Tim Ryan, to shift his research toward synaptic metabolism, encouraging a new generation of trainees from his lab and others to establish their careers in neuronal metabolism and disease. We modified this presynaptic ATP reporter by targeting it to postsynaptic (Homer2) and mitochondrial (COX8) targeting proteins to measure ATP levels in spines and mitochondria, respectively (Ghosh et al. 2025). During my postdoctoral work, I developed a genetically encoded protein synthesis inhibitor to block protein synthesis and synaptic plasticity in a single neuron, which was published in Nature Methods (Heumüller et al. 2019). The development of this reporter allows for the manipulation of structural plasticity in a cell-type-specific manner, a process widely known to be part of memory consolidation. Recently, we developed a new palette of organellar calcium sensors for the endoplasmic reticulum and mitochondria to study organellar function in neurons (Moret et al. 2025).

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